



S. No. ** 0098459 **

AIR UNIVERSITY, ISLAMABAD

ANSWER BOOK: MID SEMESTER EXAMINATION

REGISTRATION ID. 15729 CLASS & SEC B-110/CS3304 SUBJECT Artificial Intelligence
S.NO. OF EXTRA SHEET/S (IF USED) 1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
EXAMINATION (Please ☒ the appropriate Box) ☒ FALL ☐ SPRING ☐ SUMMER 2025
SIGNATURE OF STUDENT [Signature] DATE 8-Dec-2025
SIGNATURE OF INVIGILATOR [Signature] NAME OF INVIGILATOR Deepak

IMPORTANT INSTRUCTIONS

BEFORE STARTING YOUR WORK, ENSURE THAT:

1. YOU ARE IN POSSESSION OF YOUR PRINTED ROLL NUMBER SLIP AND AU ID CARD.
2. YOU ARE SITTING IN THE ROOM AND SEAT ASSIGNED TO YOU AS PER THE ROLL NUMBER SLIP.
3. YOU DO NOT HAVE ANY UNAUTHORISED PRINTED OR WRITTEN MATERIAL, CELL PHONES, SMART DEVICE OR ANY OTHER MATERIAL THAT MAY BE AN UNFAIR AID AS PER AU ACADEMIC REGULATIONS. DISCOVERY OF ANY SUCH MATERIAL AFTER START OF THE EXAMINATION WILL RENDER YOU LIABLE FOR DISCIPLINARY ACTIONS UNDER CHAPTER XV OF AU REGULATIONS.
4. WRITE YOUR REGISTRATION ID AND PARTICULARS CAREFULLY ONLY IN THE SPACES SPECIFIED ABOVE FOR THIS PURPOSE.

DURING THE EXAMINATION:

5. ENSURE THAT ANSWER BOOK AND ALL CONTINUATION SHEETS ARE SIGNED BY THE INVIGILATOR.
6. ANSWERBOOK CONTAINS 10 PAGES. TEARING OF ANY PAGE FROM THE ANSWERBOOK OR CONTINUATION SHEET IS AN OFFENCE UNDER AU UNFAIR MEANS POLICY.
7. DO NOT WRITE ANYTHING ON QUESTION PAPER EXCEPT YOUR REG ID. USE LAST PAGES OF ANSWERBOOK FOR ROUGH WORK.
8. DO NOT BORROW ANYTHING FROM OTHERS. RAISE YOUR HAND TO SEEK ASSISTANCE OF THE INVIGILATOR.
9. DO NOT USE LEAD PENCIL (EXCEPT FOR DIAGRAMS/GRAPHS ETC.) OR ANY INK OTHER THAN BLUE OR BLACK.
10. WRITE ON BOTH SIDES OF EACH PAGE IN THE ANSWERBOOK.
11. ENSURE THAT SERIAL NUMBERS OF ANSWERBOOK AND CONTINUATION SHEETS ARE ENTERED ON ATTENDANCE SHEET. ALSO ENTER SERIAL NUMBER OF ALL CONTINUATION SHEETS IN THE SPACE PROVIDED ON THE ANSWERBOOK.

AFTER THE EXAMINATION:

12. MAKE SURE THAT ALL CONTINUATION SHEETS ARE STAPLED WITH MAIN ANSWERBOOK AT THREE PLACES BEFORE SUBMITTING YOUR ANSWER SCRIPT TO THE INVIGILATOR.
13. DO NOT LEAVE EXAM HALL WITHOUT PERMISSION OF INVIGILATOR EVEN IF YOU HAVE COMPLETED THE EXAM.
14. IN NO CASE SHOULD YOU LEAVE THE EXAM HALL BEFORE HALF TIME OR YOUR PAPER MAY BE CANCELLED.

Q. NO.	1	2	3	4	5	6	7	8	9	10	TOTAL
MAXIMUM MARKS	20	30									50
MARKS OBTAINED	20	20									40

MARKS OBTAINED IN WORDS

Forty

EXAMINER'S SIGNATURE

[Signature]

Question #01

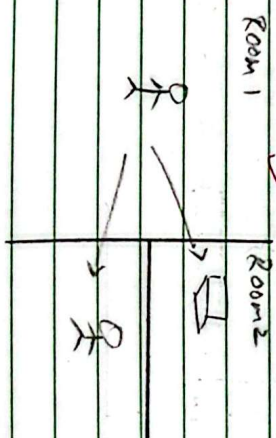
a) Artificial Intelligence :-

Artificial intelligence means to make computers to think and act like humans.

TURING TEST:-

Alan Turing (1912-1954) a British mathematician asked a question that can computer think or learn like human.

So he decided to take a test and one human is asking questions to a computer & one human in another room after the test he can't compare that which is computer answers and which is human answers.



b) AGENT :-

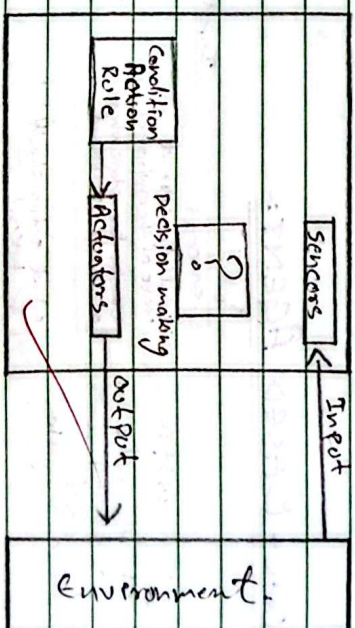
Agents are everything that perceive its environment and act upon it.

Evaluation:-

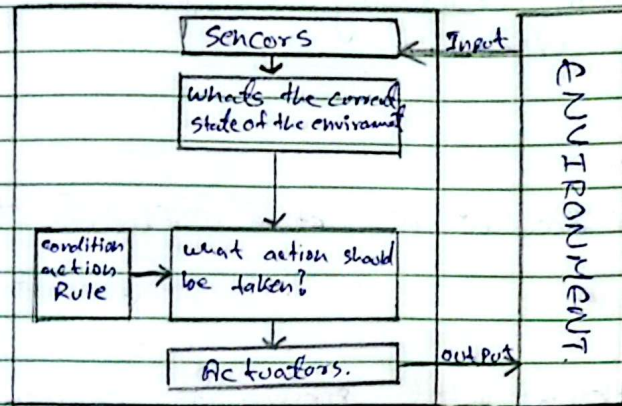
Precision is decision is action is environment

Example:-

A dust cleaning robot perceives its environment to find dust and make decision to clean it and performed action.

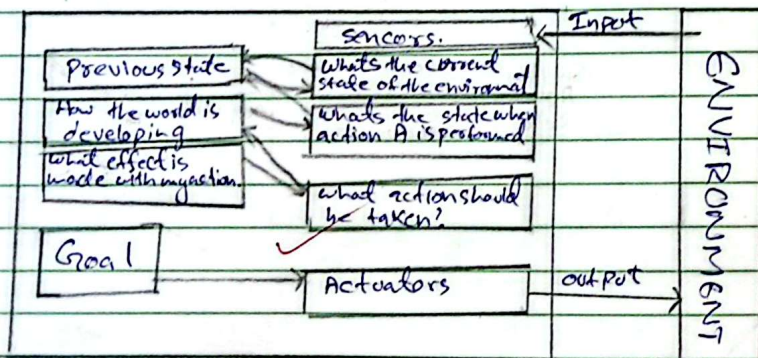


c) • SIMPLE REFLEX AGENT.



- less intelligence
- It can stuck in it loop.

• GOAL-BASED AGENT:-



- Reached to the goal without water while the path
- checks it self after performing an action.

d) PEAS MODEL.

COMPONENTS OF PEAS MODEL:-

1. Performance:-

What it can evaluate it environment

2. Environment:-

What checks the state of environment

3. Actuators:-

Act on the environment to perform a task

4. Sensors:-

Take input to give the what's the current state-

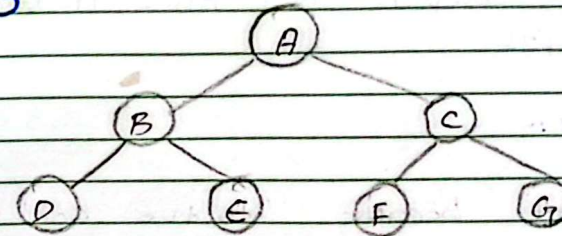
Example

A self driving car

- Performance :- fuel efficiency, time to reach at the location
- Environment :- weather condition, roads
- Actuators :- GPS, radar, cameras steering, accelerator, braking
- Sensors :- GPS, radar, cameras

Question # 02

a) DFS



15

Uses Stack data structure

1 Step 1 :- Push the A node into stack



2 Step 2 :- Pop and put its child's



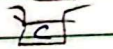
3 Step 3 :- Pop B & put its child's



4 Step 4 :- Pop D &



5 Step 5 :- Pop E



6 Step 6 :- Pop C & push its child's



7 Pop F

8 Step 8 :- Pop G

Path:-

A → B → D → E → C → F → G

Completeness:-

It is not complete some time it stuck in undefined depth.

OPTIMALITY:-

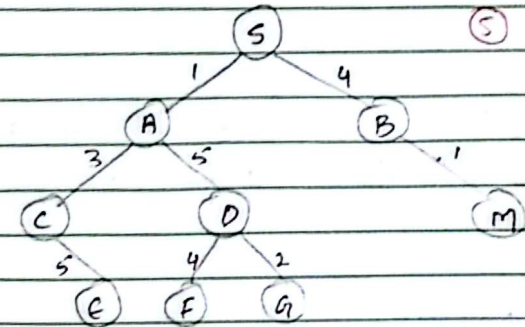
not optimal because if the goal state is on another side but it is searching on another side depth.

Time Complexity:-

$$O(b^d) = 2^2$$

$$O(4)$$

b) UCS:-



- uses queue data structure

1 The queue is arranged by cost.

[S]

[A]

[C]

[E]

[F]

[G]

Goal state found.

Total cost to reach at goal = 8.

Completeness:-

It is complete because it visits all nodes until the goal is achieved.

Optimal

Yes, it is optimal but cost increases.

Time Complexity:-

$$O(b^d)$$

$$O = 2^3$$

$$O(8)$$